iLearn Practice Exam

10BR · Chapter / Topic 1

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 1: Data Representation

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: 75

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

Questions reproduced from Cambridge International past papers © Cambridge University Press & Assessment, for classroom practice.

1 When keys are pressed on a keyboard, the text is converted to binary to be processed by the computer.

(a) Describe how the text is converted to binary to be processed by the computer.

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..... [3]

(b) Text that is input into a computer can be stored in a text file.

A text file can be compressed using lossless compression.

(i) State what effect this has on the file size.

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..... [1]

(ii) Describe how lossless compression compresses the text file.

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..... [4]

(iii) Give **two** reasons why the text file may have been compressed.

1 .....

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2 .....

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[2]

2 Humans use a denary number system and computers use a binary number system.

(a) Explain what is meant by a binary number system.

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..... [2]

(b) Convert the denary numbers 14, 59 and 234 to binary.

14 .....

59 .....

234 .....

[3]

Working space

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(c) Convert the denary numbers 9, 26 and 65 to hexadecimal.

9 .....

26 .....

65 .....

[3]

Working space

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(d) Convert the positive denary number 123 to 8-bit binary using two's complement.

Show all your working.

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..... [2]

(e) Add the binary values 00110011 and 01111000 using binary addition.

Give your answer in binary. Show all your working.

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..... [3]

3 **(a)** Hexadecimal can be used to represent binary numbers.

(i) Convert the denary number 236 into hexadecimal.

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..... [1]

Working space

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(ii) Give **one** benefit of using hexadecimal instead of binary to represent data.

.....

..... [1]

(b) Two's complement can be used to represent positive and negative binary integers.

(i) Convert the two's complement integer 10110111 into denary.

Show all your working.

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..... [2]

(ii) Convert the denary number 45 into an 8-bit two's complement integer.

.....

..... [1]

Working space

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(c) Logical shifts are performed on binary integers.

State the mathematical effect of a right shift of **four** places on a positive binary integer.

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..... [2]

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4 Data can be measured in bits.

(a) Give the name of the data storage measurement that is equal to 8 bits.

..... [1]

(b) State how many bits there are in a kibibyte (KiB).

..... [1]

(c) Give the name of the data storage measurement that is equal to 1024 gibibytes (GiB).

..... [1]

(d) A 16-bit colour image has a resolution of 512 pixels wide by 512 pixels high.

Calculate the file size of the image in kibibytes (KiB). Show all your working.

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Answer ..... KiB

[3]

5 A user enters text into a computer system, using a keyboard.

An American standard code for information interchange (ASCII) character set is used to convert the text to binary.

(a) Identify **one** other character set that could be used to convert the text to binary.

..... [1]

(b) The character ‘A’ is represented by the denary ASCII number 65.

The character ‘m’ is represented by the denary ASCII number 109.

(i) Convert the **two** denary ASCII numbers to binary.

65 .....

109 .....
[2]

Working space

.....

.....

.....

.....

(ii) Convert the **two** denary ASCII numbers to hexadecimal.

65 .....

109 .....
[2]

Working space

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(c) The character 'y' is represented by the binary ASCII number 01111001.

(i) Convert the binary ASCII number to denary.

..... [1]

Working space

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(ii) Convert the binary ASCII number to hexadecimal.

..... [1]

Working space

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(iii) A logical right shift of two places is performed on the binary ASCII number 01111001.

Give the binary number after the logical right shift of **two** places is performed.

..... [1]

Working space

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(d) The character ‘T’ is represented by the binary ASCII number 01010100.

The character ‘t’ is represented by the binary ASCII number 01110100.

Add the **two** binary numbers using binary addition. Give your answer in binary. Show all your working.

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..... [3]

6 A computer programmer uses assembly language to create a computer program for an embedded system in a washing machine.

(a) The washing machine needs to display error codes on a small screen if there is a problem with the washing machine.

The error codes are stored as binary. The binary numbers are too long to be displayed on the washing machine.

State how the error codes could be reduced in length to be displayed on the screen.

..... [1]

7 **(a)** Complete the sentences about number systems.

Use the items from the list.

Some of the items in the list will **not** be used. You should only use an item once.

A	B	C	D	E	F	G
H	W	X	Y	Z	0	1
2	4	8	10	16	127	128
255	256					

The binary number system is base ..... . The smallest denary number that can be represented as an 8-bit binary number is ..... . The largest denary number that can be represented as an 8-bit binary number is ..... .

The hexadecimal number system is base ..... . Each hexadecimal digit is equivalent to ..... bits. The numbers 1 to 9 are used and the number 10 is represented by ..... . The hexadecimal number system continues up to the number 15, which is represented by ..... .

[7]

(b) Two 8-bit binary numbers are given.

Add the two 8-bit binary numbers using binary addition.

Give your answer in binary. Show all your working.

```
  1 0 0 1 1 0 1 1
+ 0 0 0 1 0 0 1 1
```

[3]

(c) Binary addition can result in overflow.

Tick (✓) **one** box to show the correct definition of overflow in binary addition.

A The answer has created a negative number that cannot be represented in binary addition. ☐

B The answer is the result of a logical shift that cannot be performed in binary addition. ☐

C The answer is too large to represent in the number of bits available. ☐

D The answer is too small to represent in the number of bits available. ☐

[1]

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8 Two binary numbers are stored in a computer.

(a) Add the **two** binary numbers using binary addition.

Give your answer in binary. Show all of your working.

```
  11110101
+ 00111001
```

[4]

(b) A logic right shift of **three** places is performed on the binary number 11110101.

Give the denary number for the binary number that would be stored after the logical shift of **three** places has occurred. Show all of your working.

Working space .....

.....

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.....

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Denary number .....

[3]

9 Hexadecimal and binary are two examples of a number system.

(a) Give **one** other example of a number system.

..... [1]

(b) Convert the hexadecimal number 2A to an 8-bit binary number.

..... [1]

working space

.....

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(c) Convert the hexadecimal number 101 to a 12-bit binary number.

..... [1]

working space

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(d) Convert the binary number 01110001 to a hexadecimal number.

..... [1]

working space

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(e) Convert the binary number 001011011000 to a hexadecimal number.

..... [1]

working space

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(f) Add the binary numbers 10011101 and 11110000 using binary addition.

Give your answer in binary. Show all your working.

```
  1 0 0 1 1 1 0 1
+ 1 1 1 1 0 0 0 0
```

[4]

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END OF QUESTION PAPER